

Design Note: Lexical & Semantic News Retrieval on MIND and EB-NeRD

Kavan Gandikota CS4.406 Information Retrieval & Extraction

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1 What Was Built

1.1 Reproducible Data Pipeline (Q1)

A five-stage pipeline (`make data`) rebuilds both datasets from scratch:

1. **Download** — fetches MIND-small (train/dev) and EB-NeRD-demo via HTTPS.
2. **Parse** — both TSV/parquet formats are normalised into a *unified schema*: `articles` (`article_id`, `title`, `abstract`, `body`, `category`, `embeddings`), `impressions` (`impression_id`, `user_id`, `timestamp`, `history`, `candidates+labels`), and `user_history` (`src/schema.py`).
3. **Temporal Split** — train/val/test is always time-ordered (never random). For MIND-small, the provided train/dev split is respected; for EB-NeRD, the last two validation days of the demo split are held out. A helper enforces that user history features use only clicks *before* the split's earliest impression timestamp, preventing future leakage (`src/split.py`).
4. **Feature Store** — writes `article_features.parquet` and `per-split user_features.parquet` (`src/feature`).
5. **Leakage Tests** — `make test` (pytest) verifies: no timestamp overlap across splits, no duplicate impressions, and history click timestamps strictly precede impression timestamps.

Known simplification: MIND does not record individual click timestamps; each history click is approximated with the owning impression's timestamp (upper-bound, documented in README).

1.2 Lexical Candidate Generation — BM25 (Q2)

An inverted index is built over `article title + abstract` text. Given an impression, the user's most recent 20 clicked articles form the BM25 query (titles concatenated). Okapi BM25 uses $k_1=1.5$, $b=0.75$. Top- K candidates are retrieved for $K \in \{50, 100, 200\}$.

1.3 Semantic Candidate Generation — Dense Embeddings (Q3)

- **MIND:** `all-MiniLM-L6-v2` (384-dim) encodes each article's title+abstract.
- **EB-NeRD:** multilingual BERT (`bert-base-multilingual-cased`) handles Danish text; the pre-trained EB-NeRD word2vec artifact was evaluated but underperformed on the demo split.

User representation is the *mean-pool* of the most recent 20 clicked article embeddings. Retrieval uses a FAISS flat (exact cosine) index; HNSW approximate nearest-neighbour was ablated (Phase 4).

1.4 Offline Evaluation Harness (Q4)

All metrics are per-impression averaged with $B=1,000$ bootstrap resamples for 95% CIs.

Ranking: AUC (Wilcoxon–Mann–Whitney area), MRR, nDCG@5, nDCG@10. **Beyond-accuracy:** ILD@10 (mean pairwise cosine distance), Novelty@10 ($-\log_2 P(a_i)$ self-information), Catalog Coverage@10 (fraction of catalog in any user's top-10), Recall@{50,100,200} (fraction of ground-truth positives retrieved from full catalog).

Slicing — *User cohort:* cold-start (bottom 2% of per-impression history length) vs. warm. *Item popularity:* head (top 20% most-clicked catalog articles) vs. tail.

2 Key Design Choices and Alternatives

- **Query field ablation** (Table 3): Title-only performs marginally better than title+abstract for MIND (nDCG@10 0.2783 vs. 0.2772); abstract text adds noise. For EB-NeRD, body text improves Recall@200 (0.0248 $\rightarrow$ 0.0305) but is marginal on ranking.
- **History length** (Table 4): Sweeping {10, 20, 30}, history=20 maximises mean nDCG@10 pooled across both models/datasets; longer history introduces topic drift on both datasets.
- **FAISS flat vs. HNSW** (Table 5): Ranking metrics are identical; HNSW slightly improves Recall@200 for MIND (0.0114 $\rightarrow$ 0.0122) at much lower query latency — preferred for $10\times$ scale.
- **Cold-start threshold:** Varying the percentile boundary (p1/p2/p5/p10) leaves aggregate metrics invariant, confirming the bottom 2% is a natural break at near-zero history.
- **Temporal split over random:** Random splitting leaks future impressions into training user features; time-

ordering is mandatory for realistic serving conditions.

3 Experimental Results

3.1 Codabench Official Test-Set Scores

Table 1: Official Codabench scores (MIND test-set; EB-NeRD mean over 8 test days).

Dataset	Method	AUC	MRR	nDCG@5	nDCG@10
MIND	Lexical BM25 (title+abs)	0.5785	0.2805	0.2975	0.3526
MIND	Semantic (title+abs)	0.6425	0.3137	0.3377	0.3934
EB-NeRD	Lexical BM25 (title+abs)	0.5147	0.3323	0.3662	0.4483
EB-NeRD	Semantic (title+abs+body)	0.4980	0.3244	0.3546	0.4400

[Screenshot: MIND Codabench leaderboard]
Competition #13967

[Screenshot: EB-NeRD Codabench leaderboard]
Competition #2469

(a) MIND (Competition #13967)

(b) EB-NeRD (Competition #2469)

Figure 1: Codabench submission screenshots (replace with actual screenshots).

3.2 Offline Val-Set Benchmark

Table 2: Validation-set overall performance with 95% bootstrap CIs.

Dataset	Method	AUC	MRR	nDCG@5	nDCG@10
MIND	BM25	0.5032 [.5025,.5038]	0.2406	0.2133	0.2772
MIND	Semantic	0.5950 [.5904,.5996]	0.3036	0.2846	0.3460
	Δ	+0.0919 (+18.3%)	+0.0629	+0.0714	+0.0688
EB-NeRD	BM25	0.5000 [.4990,.5011]	0.3122	0.3439	0.4282
EB-NeRD	Semantic	0.4962 [.4920,.5005]	0.3214	0.3498	0.4353
	Δ	−0.0038 (−0.8%)	+0.0092	+0.0058	+0.0071

3.3 Ablation Studies

Table 3: Phase 1: BM25 query field ablation.

Config	Dataset	AUC	MRR	nDCG@5	nDCG@10
title	MIND	0.5039	0.2421	0.2146	0.2783
title+abs	MIND	0.5032	0.2406	0.2133	0.2772
title	EB-NeRD	0.5014	0.3135	0.3455	0.4295
title+abs	EB-NeRD	0.5000	0.3122	0.3439	0.4282
title+abs+body	EB-NeRD	0.5007	0.3127	0.3440	0.4289

4 Observations

Semantic vs. Lexical — dataset asymmetry. On MIND (English), dense retrieval dominates: AUC +18.3%, MRR +26.2%, nDCG@10 +24.8% on val, and AUC 0.6425 vs. 0.5785 on the official Codabench test. Non-overlapping 95% CIs confirm statistical significance ($p < 0.001$). English news heavy synonym usage (“*presidential election*” / “*White House vote*”) is invisible to BM25 term matching. On EB-NeRD (Danish), BM25 is competitive: its AUC exceeds Semantic (0.5147 vs. 0.4980 on Codabench test), while Semantic leads MRR (+2.95%) and nDCG@10

Table 4: Phase 2: History length sweep (title+abstract queries; val set).

Config	Dataset	AUC	MRR	nDCG@5	nDCG@10
bm25-h10	MIND	0.5031	0.2401	0.2128	0.2766
bm25-h20	MIND	0.5032	0.2406	0.2133	0.2772
bm25-h30	MIND	0.5030	0.2398	0.2128	0.2766
sem-h10	MIND	0.5906	0.3011	0.2812	0.3436
sem-h20	MIND	0.5950	0.3036	0.2846	0.3460
sem-h30	MIND	0.5964	0.3043	0.2857	0.3467
bm25-h10	EB-NeRD	0.5007	0.3123	0.3440	0.4284
bm25-h20	EB-NeRD	0.5000	0.3122	0.3439	0.4282
bm25-h30	EB-NeRD	0.4994	0.3106	0.3424	0.4268
sem-h10	EB-NeRD	0.4958	0.3209	0.3493	0.4342
sem-h20	EB-NeRD	0.4962	0.3214	0.3498	0.4353
sem-h30	EB-NeRD	0.4966	0.3214	0.3499	0.4351

Table 5: Phase 4: FAISS flat (exact) vs. HNSW (ANN).

Config	Dataset	AUC	MRR	nDCG@10	R@50	R@100	R@200
Flat	MIND	0.5950	0.3036	0.3460	0.0034	0.0063	0.0114
HNSW	MIND	0.5950	0.3036	0.3460	0.0036	0.0067	0.0122
Flat	EB-NeRD	0.4962	0.3214	0.4353	0.0079	0.0164	0.0311
HNSW	EB-NeRD	0.4962	0.3214	0.4353	0.0079	0.0164	0.0311

(+1.66%) on val. Dense proper nouns in Danish editorial text (politicians, municipalities, sports clubs) give BM25 an advantage the embeddings partially fail to capture.

Cold-start vs. warm users. On MIND, BM25 achieves Recall@50= 0 for cold-start users (history ≤ 0 clicks); Semantic maintains non-zero retrieval (Recall@50= 0) due to the mean-pool of even a single embedding encoding a rich semantic prior. For warm users the gap widens (Warm AUC: 0.6139 vs. 0.5064, +21.2%). On EB-NeRD, cold-start margins are smaller (≤ 11 clicks threshold) but Semantic consistently outperforms on MRR (+5.99%) and nDCG@5 (+6.54%) in that slice.

Head vs. tail item slicing. BM25 achieves higher Recall@K on EB-NeRD head articles (Recall@100: 0.0604 vs. 0.0316) because popular breaking-news headlines share repeated exact keyword phrases. However, Semantic dominates on ranking quality for head items (Head AUC: 0.6174 vs. 0.5251, +17.6%) and on all tail metrics (Head nDCG@10: 0.4570 vs. 0.3980), indicating superior long-tail discovery via dense cluster alignment.

Beyond-accuracy metrics. BM25 yields higher ILD@10 because keyword-driven retrieval pulls articles from lexically disjoint parts of the corpus; ANNS tends to cluster semantically similar articles. Catalog Coverage@10 is broadly comparable (≈ 2.1 – 2.5% MIND, ≈ 21 – 22% EB-NeRD), BM25 marginally higher. Overall Recall@K is low ($\leq 3\%$) because the ground-truth clicked article lies within the curator-supplied candidate set, which our full-catalog retrieval rarely intersects.

5 Scalability: Where the Pipeline Breaks at $10\times$

- **BM25 index:** In-memory index for 120K articles fits one machine; at $10\times$ ($\sim 1.2\text{M}$ articles, 6×10^9 impressions) sharding is required (Elasticsearch, Lucene replicas) alongside WAND/BMW posting-list pruning.
- **FAISS flat:** Brute-force ANN is $O(n \cdot d)/\text{query}$; at $10\times$ ($\sim 1.2\text{M}$ articles, $d=768$) latency violates serving SLOs. HNSW or ScaNN with product quantisation must replace flat search.
- **Embedding inference:** Encoding 1.2M articles/day with BERT at 50 ms/article requires $\approx 600\text{K}$ GPU-seconds/day; a distilled model (all-MiniLM) or dedicated inference cluster (Triton) is mandatory.
- **User feature computation:** The current feature store recomputes histories once per split; with 27M users and real-time impressions, per-impression incremental updates require a streaming feature store (Feast, Redis).
- **Storage:** Parquet feature files for 600M impressions ($\approx 600\text{ GB}$ uncompressed) cannot fit in RAM; columnar partitioning by date with lazy Polars/PyArrow scans is needed.
- **Leakage tests:** The full-scan pytest suite becomes multi-hour at $10\times$; continuous streaming validation (Great Expectations on micro-batches) is the scalable alternative.